

Rust Bobble Head Simulation

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November 1, 2025

1 Introduction

For this project I plan on simulating a bobble heads within in Rust, like how some people have a bobble head for their car's dashboard I plan to a GUI rust equivalent but for the PC desktop. It is planned to have QML for the visuals while it utilizes rust for the actual simulation numerical simulation. The user will be able to drag on the head to different degrees of offsets and then watch it bobble.



Figure 1: [Bobblehead on Wikipedia](#)

2 Specifications, Issues And Concerns

QT's QML doesn't yet have official Rust bindings so there might be issues with trying to get it run in a clean manner. Also QML single and slots is very closely tied to how C++ handles its pointers and memory management. I hope this doesn't mean that it will result in unsafe Rust but I seen other examples of QML and Rust being linked together. They've attested to the safety of using a Rust backend.

2.1 Planned Basic Features

2.1.1 Draggable Head

The user will be able to mouse over the bobble head on the desktop causing it to bobble. They are the able to click drag on the bobble head to pull on it to make bobble to a greater degree.

2.1.2 Different Types of Bobble Heads

The Rust crab (Ferris the Rustacean), Cat, Baseball Player.

2.1.3 Cross Platform Support

The bobble heads application will be able to run on Windows and Linux.

3 GitHub Link

Project Link: <https://github.com/FelixFalling/rustBobbleHeadSimulation>.